# Question B1

This question carries 17 marks in total.

**(a) [5 MARKS]** Consider the conical surface, $S$, defined by

$$z = 4\sqrt{x^2+y^2} \quad \text{for} \quad -1 \le x, y \le 1.$$

Show that the normal to $S$, pointing *out* of the cone, can be written as:

$$\mathbf{n} = \frac{4x}{\sqrt{x^2+y^2}}\mathbf{i} + \frac{4y}{\sqrt{x^2+y^2}}\mathbf{j} - \mathbf{k}.$$

**(b) [3 MARKS]** Given the same surface as in part (a), show that $S$ can be parameterized as follows:

$$\mathbf{r}(s,t) = s\cos(t)\mathbf{i} + s\sin(t)\mathbf{j} + 4s\mathbf{k}, \quad \text{for} \quad 0 \le s \le 1, 0 \le t \le 2\pi.$$

**(c) [3 MARKS]** Let

$$\mathbf{F} = y\mathbf{i} - x\mathbf{j} + 5\mathbf{k}.$$

Calculate the flux of $\mathbf{F}$ *out* of the cone. i.e. evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $S$ is the same surface as in part (a) and $d\mathbf{S}$ is the vector element of area pointing out of the cone.

**(d) [6 MARKS]** Let $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + 5t\mathbf{k}$ define a helix $C$ for $0 \le t \le 2\pi$.

Calculate

$$\int_C \mathbf{F} \cdot d\mathbf{r}, \quad \text{for} \quad \mathbf{F} = z\mathbf{i} - x\mathbf{j} + y\mathbf{k}.$$

You may find the identity $\cos^2(A) = \frac{1}{2}(1+\cos(2A))$ useful.